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Automated IPAM & BoQ Project: Implementing Multi-VLAN DHCP with Active Port Security

NAMA : KEYSA ANNAJWA

KELAS : XI SMK TKJ

**TUJUAN : Melatih integrasi data inventaris jaringan (BoQ),
segmentasi keamanan (VLAN), dan otomatisasi pemetaan IP Address**

**TOOLS/RUMUS : =COUNTA(), =SUM(),VLOOKUP, DATA VALIDATION,
Cisco Packet Tracer (CLI),**

1. HOST SPECIFICATION & OS DISTRIBUTION MATRIX

DEVICE DATA ALL

N0	Device ID	Device Name	Brand	Model	OS Version	Location (Floor 1 / 2)	Unit price
1	PC-1	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
2	PC-2	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
3	PC-3	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
4	PC-4	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
5	PC-5	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
6	PC-6	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
7	PC-7	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
8	PC-8	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
9	PC-9	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
10	PC-10	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
11	PC-11	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
12	PC-12	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
13	PC-13	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
14	PC-14	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
15	PC-15	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
16	PC-16	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
17	PC-17	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
18	PC-18	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
19	PC-19	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
20	PC-20	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	Rp 1.150.000
21	PC-21	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
22	PC-22	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
23	PC-23	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
24	PC-24	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
25	PC-25	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
26	PC-26	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
27	PC-27	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
28	PC-28	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
29	PC-29	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
30	PC-30	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
31	PC-31	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
32	PC-32	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
33	PC-33	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
34	PC-34	PC	(Custom Brand)	Intel Core i5-4570/4591	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
35	PC-35	PC	(Custom Brand)	Intel Core i5-4570/4592	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
36	PC-36	PC	(Custom Brand)	Intel Core i5-4570/4593	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
37	PC-37	PC	(Custom Brand)	Intel Core i5-4570/4594	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
38	PC-38	PC	(Custom Brand)	Intel Core i5-4570/4595	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	Rp 1.150.000
39	Switch F-CS	SWITCH	MIKROTIK	MikroTik CRS326-24G-2S+RM (24 Port)	SwOS	FLOOR 1 OPERATIONAL	Rp 3.365.000
40	Switch IT-MA	SWITCH	MIKROTIK	MikroTik CRS326-24G-2S+RM (24 Port)	SwOS	FLOOR 2 MANAGEMENT & TECH	Rp 3.365.001
41	Router-U	ROUTER	MIKROTIK	MikroTik hEX S (RB760iGS) (5 port)	ROUTERoS v7	FLOOR 2 MANAGEMENT & TECH	Rp 1.100.000
42	Switch-U	SWITCH	MIKROTIK	MikroTik CRS304-4XG-IN	ROUTERoS	FLOOR 2 MANAGEMENT & TECH	Rp 3.940.000

2. AUTOMATED PORT & IP MAPPING MATRIX (FLOOR 1 & 2)

Automated Port & IP Mapping

NO	Device ID	Device OWN	VLAN ID	IP Address	Interface	Connected To	Status
1	PC-1	CS	VLAN 10	192.168.10.2	FastEthernet0/1	SWITCH F-CS-RT	ACTIVE
2	PC-2	CS	VLAN 10	192.168.10.3	FastEthernet0/2	SWITCH F-CS-RT	ACTIVE
3	PC-3	CS	VLAN 10	192.168.10.4	FastEthernet0/3	SWITCH F-CS-RT	ACTIVE
4	PC-4	CS	VLAN 10	192.168.10.5	FastEthernet0/4	SWITCH F-CS-RT	ACTIVE
5	PC-5	CS	VLAN 10	192.168.10.6	FastEthernet0/5	SWITCH F-CS-RT	ACTIVE
6	PC-6	RECEPTIONIST	VLAN 20	192.168.20.2	FastEthernet0/6	SWITCH F-CS-RT	ACTIVE
7	PC-7	RECEPTIONIST	VLAN 20	192.168.20.3	FastEthernet0/7	SWITCH F-CS-RT	ACTIVE
8	PC-8	RECEPTIONIST	VLAN 20	192.168.20.4	FastEthernet0/8	SWITCH F-CS-RT	ACTIVE
9	PC-9	RECEPTIONIST	VLAN 20	192.168.20.5	FastEthernet0/9	SWITCH F-CS-RT	ACTIVE
10	PC-10	RECEPTIONIST	VLAN 20	192.168.20.6	FastEthernet0/10	SWITCH F-CS-RT	ACTIVE
11	PC-11	FINANCE	VLAN 30	192.168.30.2	FastEthernet0/11	SWITCH F-CS-RT	ACTIVE
12	PC-12	FINANCE	VLAN 30	192.168.30.3	FastEthernet0/12	SWITCH F-CS-RT	ACTIVE
13	PC-13	FINANCE	VLAN 30	192.168.30.4	FastEthernet0/13	SWITCH F-CS-RT	ACTIVE
14	PC-14	FINANCE	VLAN 30	192.168.30.5	FastEthernet0/14	SWITCH F-CS-RT	ACTIVE
15	PC-15	FINANCE	VLAN 30	192.168.30.6	FastEthernet0/15	SWITCH F-CS-RT	ACTIVE
16	PC-16	FINANCE	VLAN 30	192.168.30.7	FastEthernet0/16	SWITCH F-CS-RT	ACTIVE
17	PC-17	FINANCE	VLAN 30	192.168.30.8	FastEthernet0/17	SWITCH F-CS-RT	ACTIVE
18	PC-18	FINANCE	VLAN 30	192.168.30.9	FastEthernet0/18	SWITCH F-CS-RT	ACTIVE
19	PC-19	FINANCE	VLAN 30	192.168.30.10	FastEthernet0/19	SWITCH F-CS-RT	ACTIVE
20	PC-20	FINANCE	VLAN 30	192.168.30.11	FastEthernet0/20	SWITCH F-CS-RT	ACTIVE
21	PC-21	MANAGER	VLAN 40	192.168.40.2	FastEthernet0/1	SWITCH IT-MA	ACTIVE
22	PC-22	MANAGER	VLAN 40	192.168.40.3	FastEthernet0/2	SWITCH IT-MA	ACTIVE
23	PC-23	MANAGER	VLAN 40	192.168.40.4	FastEthernet0/3	SWITCH IT-MA	ACTIVE
24	PC-24	IT ENGINEEER	VLAN 50	192.168.50.2	FastEthernet0/4	SWITCH IT-MA	ACTIVE
25	PC-25	IT ENGINEEER	VLAN 50	192.168.50.3	FastEthernet0/5	SWITCH IT-MA	ACTIVE
26	PC-26	IT ENGINEEER	VLAN 50	192.168.50.4	FastEthernet0/6	SWITCH IT-MA	ACTIVE
27	PC-27	IT ENGINEEER	VLAN 50	192.168.50.5	FastEthernet0/7	SWITCH IT-MA	ACTIVE
28	PC-28	IT ENGINEEER	VLAN 50	192.168.50.6	FastEthernet0/8	SWITCH IT-MA	ACTIVE
29	PC-29	IT ENGINEEER	VLAN 50	192.168.50.7	FastEthernet0/9	SWITCH IT-MA	ACTIVE
30	PC-30	IT ENGINEEER	VLAN 50	192.168.50.8	FastEthernet0/10	SWITCH IT-MA	ACTIVE
31	PC-31	IT ENGINEEER	VLAN 50	192.168.50.9	FastEthernet0/11	SWITCH IT-MA	ACTIVE
32	PC-32	IT ENGINEEER	VLAN 50	192.168.50.10	FastEthernet0/12	SWITCH IT-MA	ACTIVE
33	PC-33	IT ENGINEEER	VLAN 50	192.168.50.11	FastEthernet0/13	SWITCH IT-MA	ACTIVE
34	PC-34	IT ENGINEEER	VLAN 50	192.168.50.12	FastEthernet0/14	SWITCH IT-MA	ACTIVE
35	PC-35	IT ENGINEEER	VLAN 50	192.168.50.13	FastEthernet0/15	SWITCH IT-MA	ACTIVE
36	PC-36	IT ENGINEEER	VLAN 50	192.168.50.14	FastEthernet0/16	SWITCH IT-MA	ACTIVE
37	PC-37	IT ENGINEEER	VLAN 50	192.168.50.15	FastEthernet0/17	SWITCH IT-MA	ACTIVE
38	PC-38	IT ENGINEEER	VLAN 50	192.168.50.16	FastEthernet0/18	SWITCH IT-MA	ACTIVE
39	Switch-FCS	CS & RECEPTIONIS			FastEthernet0/21	SWITCH-U	ACTIVE
40					GigabitEthernet1/0	ROUTER-U	ACTIVE
41	Switch-IT_MA	IT ENGINEEER			FastEthernet0/19	SWITCH-U	ACTIVE
					GigabitEthernet0/1	ROUTER-U	ACTIVE
42	Router-U	ALL			GigabitEthernet3/0	SWITCH IT-MA	ACTIVE
					GigabitEthernet6/0	SWITCH F-CS-RT	ACTIVE
43	Switch-U	ALL			FastEthernet0/1	SWITCH F-CS-RT	ACTIVE
					FastEthernet1/1	SWITCH IT-MA	ACTIVE

3. CONFIGURATION VERIFICATION & SECURITY STATUS

FLOOR 1 - CS & RECEPTIONIST & FINANCE INFRASTRUCTURE					
NO	Device	Verification Parameter	Status	CLI Output	Evidence File Name
1	Switch F-CS-RT	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 10 Interface VLAN 20 Interface VLAN 30 Interface TRUNK	F.1-P.1
2	Router-U	IP DHCP POOL ACCESS	ACTIVE	IP DHCP POOL VLAN 10 IP DHCP POOL VLAN 20 IP DHCP POOL VLAN 30 IP DHCP POOL VLAN 40 IP DHCP POOL VLAN 50	F.1-P.2
3	Switch F-CS-RT	`Enable Password `line console Password `Vty line Password `Encryption Password `Banner MOTD	ACTIVE	enable secret ofliCeEnFCS line console password OFHcEiC\$t line Vty password fC\$tYOf Banner MOTD	F.1-P.3
4	Switch F-CS-RT	Port-Security	ACTIVE	Port security Fa0/9-18	F.1-P.4

FLOOR 2 - IT ENGINEER & MANAGER INFRASTRUCTURE					
NO	Device	Verification Parameter	Status	CLI Output	Evidence File Name
1	Switch IT-MA	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 40 Interface VLAN 50 Interface TRUNK	F.2-P.1
3	Switch IT-MA	`Enable Password `line console Password `Vty line Password `Encryption Password `Banner MOTD	ACTIVE	enable secret AkT1vsW5\$c line console password ltMaNfL00r1 line Vty password FI0oRlitma Banner MOTD	F.2-P.2

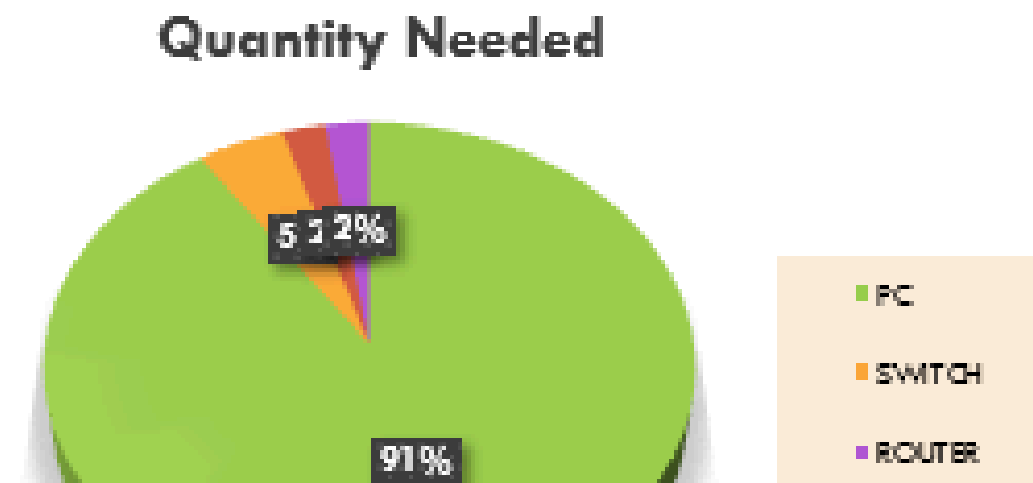
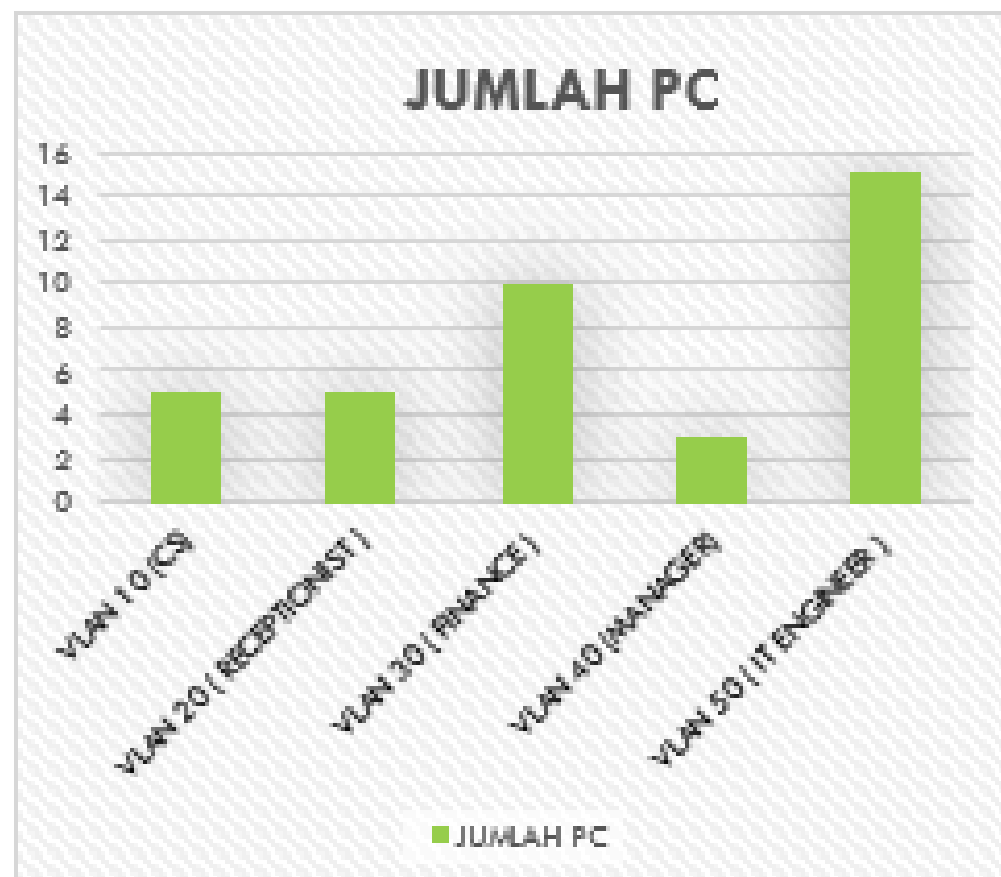
4. BILL OF MATERIALS (BoQ) & INFRASTRUCTURE BUDGETING

NO	Device Name	Quantity Needed	Device Model	Unit Price	Total price
1	PC	38	Intel Core i5-4570/4590	Rp 1.500.000	Rp 43.700.000
2	SWITCH	2	MikroTik CRS326-24G-2S+RM (24 Port)	Rp 3.365.000	Rp 6.730.001
3		1	MikroTik hEX S (RB760iGS) (5 port)	Rp 1.100.000	Rp 1.100.000
4	ROUTER	1	MikroTik CRS304-4XG-IN	Rp 3.940.000	Rp 3.940.000
				Total Price	Rp 55.470.001

5. ENTERPRISE NETWORK DASHBOARD: VLAN & COST ANALYSIS

VLAN ID	JUMLAH PC
VLAN 10	5
VLAN 20	5
VLAN 30	10
VLAN 40	3
VLAN 50	15

Device	Quantity Needed	Unit Price	Total price
PC	38	Rp 1.500.000	Rp 43.700.000
SWITCH	2	Rp 3.365.000	Rp 6.730.001
	1	Rp 1.100.000	Rp 1.100.000
ROUTER	1	Rp 3.940.000	Rp 3.940.000



Automated IPAM & BoQ Project: Implementing Multi-VLAN DHCP with Active Port Security

NAMA :KEYSA ANNAJWA

KELAS :XI SMK TKJ

TUJUAN :Melatih integrasidatainventarisjaringan(BoQ),
segmentasi keamanan (VLAN), dan otomasi pemetaan IP Address
TOOLS/RUMUS:=COUNT(), =SUM(),VLOOKUP, DATA VALIDATION, Cisco
PacketTracer (CLI),

1. HOST SPECIFICATION & OS DISTRIBUTION MATRIX

DEVICE DATA ALL							
NO	Device ID	Device Name	B ran d	Mo d el	OS Version	Location (Floor 1 / 2)	Unit price
1	PC-1	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
2	PC-2	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
3	PC-3	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
4	PC-4	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
5	PC-5	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
6	PC-6	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
7	PC-7	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
8	PC-7	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
9	PC-8	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
10	PC-9	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
11	PC-10	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,190,000
12	PC-11	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
13	PC-12	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
14	PC-13	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
15	PC-14	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
16	PC-15	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
17	PC-16	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
18	PC-16	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
19	PC-17	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
20	PC-18	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
21	PC-19	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
22	PC-19	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
23	PC-20	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 1 OPERATIONAL	1,150,000
24	PC-21	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
25	PC-22	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
26	PC-23	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
27	PC-24	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
28	PC-25	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
29	PC-26	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
30	PC-27	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
31	PC-27	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
32	PC-28	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
33	PC-29	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
34	PC-30	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
35	PC-31	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
36	PC-32	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
37	PC-33	PC	(Custom Brand)	Intel Core i5-4570/4590	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
38	PC-34	PC	(Custom Brand)	Intel Core i5-4570/4591	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,190,000
39	PC-35	PC	(Custom Brand)	Intel Core i5-4570/4592	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,190,000
40	PC-36	PC	(Custom Brand)	Intel Core i5-4570/4593	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,190,000
41	PC-37	PC	(Custom Brand)	Intel Core i5-4570/4594	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
42	PC-38	PC	(Custom Brand)	Intel Core i5-4570/4595	Windows 10 Pro 64-bit	FLOOR 2 MANAGEMENT & TECH	1,150,000
43	Switch-FCS	MIKROTIK	MIKROTIK	MIKROTIK CRS226-24G-26-RM (24 Port)	swOS	FLOOR 2 MANAGEMENT & TECH	2,345,000
44	Switch-IT-MA	SWITCH	MIKROTIK	MIKROTIK CRS226-24G-26-RM (24 Port)	swOS	FLOOR 2 MANAGEMENT & TECH	2,345,000
45	Router-U	SWITCH	MIKROTIK	MIKROTIK REX-5 (RB760IGS) (5 port)	ROUTERs v7	FLOOR 2 MANAGEMENT & TECH	1,600,000
46	Switch-U	ROUTER	MIKROTIK	MIKROTIK CRS304-4XG-1N	ROUTERs	FLOOR 2 MANAGEMENT & TECH	3,040,000
47		SWITCH					

2. AUTOMATED PORT & IP MAPPING MATRIX (FLOOR 1 & 2)

Automated Port & IP Mapping							
NO	Device ID	Device OWN	VLAN ID	IP Address	Interface	Connected To	Status
1	PC-1	CS	VLAN 10	192.168.10.2	FastEthernet0/1	SWITCH F-CS-RY	ACTIVE
2	PC-2	CS	VLAN 10	192.168.10.3	FastEthernet0/2	SWITCH F-CS-RY	ACTIVE
3	PC-3	CS	VLAN 10	192.168.10.4	FastEthernet0/3	SWITCH F-CS-RY	ACTIVE
4	PC-4	CS	VLAN 10	192.168.10.5	FastEthernet0/4	SWITCH F-CS-RY	ACTIVE
5	PC-5	CS	VLAN 10	192.168.10.6	FastEthernet0/5	SWITCH F-CS-RY	ACTIVE
6	PC-6	RECEPTIONIST	VLAN 20	192.168.20.2	FastEthernet0/6	SWITCH F-CS-RY	ACTIVE
7	PC-7	RECEPTIONIST	VLAN 20	192.168.20.3	FastEthernet0/7	SWITCH F-CS-RY	ACTIVE
8	PC-8	RECEPTIONIST	VLAN 20	192.168.20.4	FastEthernet0/8	SWITCH F-CS-RY	ACTIVE
9	PC-9	RECEPTIONIST	VLAN 20	192.168.20.5	FastEthernet0/9	SWITCH F-CS-RY	ACTIVE
10	PC-10	RECEPTIONIST	VLAN 20	192.168.20.6	FastEthernet0/10	SWITCH F-CS-RY	ACTIVE
11	PC-11	FINANCE	VLAN 30	192.168.30.2	FastEthernet0/11	SWITCH F-CS-RY	ACTIVE
12	PC-12	FINANCE	VLAN 30	192.168.30.3	FastEthernet0/12	SWITCH F-CS-RY	ACTIVE
13	PC-13	FINANCE	VLAN 30	192.168.30.4	FastEthernet0/13	SWITCH F-CS-RY	ACTIVE
14	PC-14	FINANCE	VLAN 30	192.168.30.5	FastEthernet0/14	SWITCH F-CS-RY	ACTIVE
15	PC-15	FINANCE	VLAN 30	192.168.30.6	FastEthernet0/15	SWITCH F-CS-RY	ACTIVE
16	PC-16	FINANCE	VLAN 30	192.168.30.7	FastEthernet0/16	SWITCH F-CS-RY	ACTIVE
17	PC-17	FINANCE	VLAN 30	192.168.30.8	FastEthernet0/17	SWITCH F-CS-RY	ACTIVE
18	PC-18	FINANCE	VLAN 30	192.168.30.9	FastEthernet0/18	SWITCH F-CS-RY	ACTIVE
19	PC-19	FINANCE	VLAN 30	192.168.30.10	FastEthernet0/19	SWITCH F-CS-RY	ACTIVE
20	PC-20	FINANCE	VLAN 30	192.168.30.11	FastEthernet0/20	SWITCH F-CS-RY	ACTIVE
21	PC-21	MANAGER	VLAN 40	192.168.40.2	FastEthernet0/1	SWITCH IT-MA	ACTIVE
22	PC-22	MANAGER	VLAN 40	192.168.40.3	FastEthernet0/2	SWITCH IT-MA	ACTIVE
23	PC-23	MANAGER	VLAN 40	192.168.40.4	FastEthernet0/3	SWITCH IT-MA	ACTIVE
24	PC-24	IT ENGINEER	VLAN 50	192.168.50.2	FastEthernet0/4	SWITCH IT-MA	ACTIVE
25	PC-25	IT ENGINEER	VLAN 50	192.168.50.3	FastEthernet0/5	SWITCH IT-MA	ACTIVE
26	PC-26	IT ENGINEER	VLAN 50	192.168.50.4	FastEthernet0/6	SWITCH IT-MA	ACTIVE
27	PC-27	IT ENGINEER	VLAN 50	192.168.50.5	FastEthernet0/7	SWITCH IT-MA	ACTIVE
28	PC-28	IT ENGINEER	VLAN 50	192.168.50.6	FastEthernet0/8	SWITCH IT-MA	ACTIVE
29	PC-29	IT ENGINEER	VLAN 50	192.168.50.7	FastEthernet0/9	SWITCH IT-MA	ACTIVE
30	PC-30	IT ENGINEER	VLAN 50	192.168.50.8	FastEthernet0/10	SWITCH IT-MA	ACTIVE
31	PC-31	IT ENGINEER	VLAN 50	192.168.50.9	FastEthernet0/11	SWITCH IT-MA	ACTIVE
32	PC-32	IT ENGINEER	VLAN 50	192.168.50.10	FastEthernet0/12	SWITCH IT-MA	ACTIVE
33	PC-33	IT ENGINEER	VLAN 50	192.168.50.11	FastEthernet0/13	SWITCH IT-MA	ACTIVE
34	PC-34	IT ENGINEER	VLAN 50	192.168.50.12	FastEthernet0/14	SWITCH IT-MA	ACTIVE
35	PC-35	IT ENGINEER	VLAN 50	192.168.50.13	FastEthernet0/15	SWITCH IT-MA	ACTIVE
36	PC-36	IT ENGINEER	VLAN 50	192.168.50.14	FastEthernet0/16	SWITCH IT-MA	ACTIVE
37	PC-37	IT ENGINEER	VLAN 50	192.168.50.15	FastEthernet0/17	SWITCH IT-MA	ACTIVE
38	PC-38	IT ENGINEER	VLAN 50	192.168.50.16	FastEthernet0/18	SWITCH IT-MA	ACTIVE
39	Switch-FCS	CS & RECEPTIONIST	*****		FastEthernet0/21	ROUTER-U	ACTIVE
40						ROUTER-U	ACTIVE
41					GigabitEthernet0/19	ROUTER-U	ACTIVE
	Switch-IT_MA	IT ENGINEER	*****		FastEthernet0/20	SWITCH IT-MA	ACTIVE
					GigabitEthernet0/17	SWITCH F-CS-RY	ACTIVE
42	Router-U	ALL	*****		GigabitEthernet0/0	SWITCH IT-MA	ACTIVE
43	Switch-U	ALL	*****		FastEthernet0/1		
					FastEthernet0/1		

3. CONFIGURATION VERIFICATION & SECURITY STATUS

FLOOR 1 - CS & RECEPTIONIST & FINANCE INFRASTRUCTURE					
NO	Device	Verification	Status	CLI Output	Evidence
1	Switch F-CS-RT	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 10 Interface VLAN 20 Interface VLAN 30 Interface TRUNK	F.1-P.1
2	Router-U	IP DHCP POOL ACCESS	ACTIVE	IP DHCP POOL VLAN 10 IP DHCP POOL VLAN 20 IP DHCP POOL VLAN 30 IP DHCP POOL VLAN 40 IP DHCP POOL VLAN 50	F.1-P.2
3	Switch F-CS-RT	` Enable Password ` line console Password ` Vty line Password ` Encryption Password ` Banner MOTD	ACTIVE	enable secret of1CeEnFCS line console password OFH cEiC \$ T line Vty password iC\$TtOFH Banner MOTD	F.1-P.3
4	Switch F-CS-RT	Port-Security	ACTIVE	Port security Fa0/9-18	F.1-P.4

FLOOR 2 - IT ENGINEER & MANAGER INFRASTRUCTURE					
NO	Device	Verification Parameter	Status	CLI Output	Evidence File Name
1	Switch IT-MA	VLAN ACCESS TRUNK ACCESS	ACTIVE	Interface VLAN 40 Interface VLAN 50 Interface TRUNK	F.2-P.1
3	Switch IT-MA	` Enable Password ` line console Password ` vty line Password ` Encryption Password ` Banner MOTD	ACTIVE	enable secret A K T 1 v s W 5 \$ c line console password ! ! Ma N R L 0 0 r 1 line Vty password F l o o R i t m a Banner MOTD	F.2-P.2

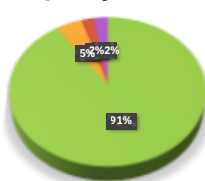
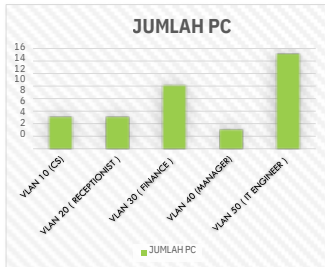
4. BILL OF MATERIALS (BoQ) & INFRASTRUCTURE BUDGETING

1	PC	38	Intel Core i5-4570/4590	Rp	1.500.000	Rp	43.700.000
2	SWITCH	2	MikroTik CRS326-24G-2S+RM (24 Port)				
4	ROUTER	1	MikroTik CRS304-4XG-IN	Rp	3.940.000	R	3.940.000
				Total Price		R	55.470.001

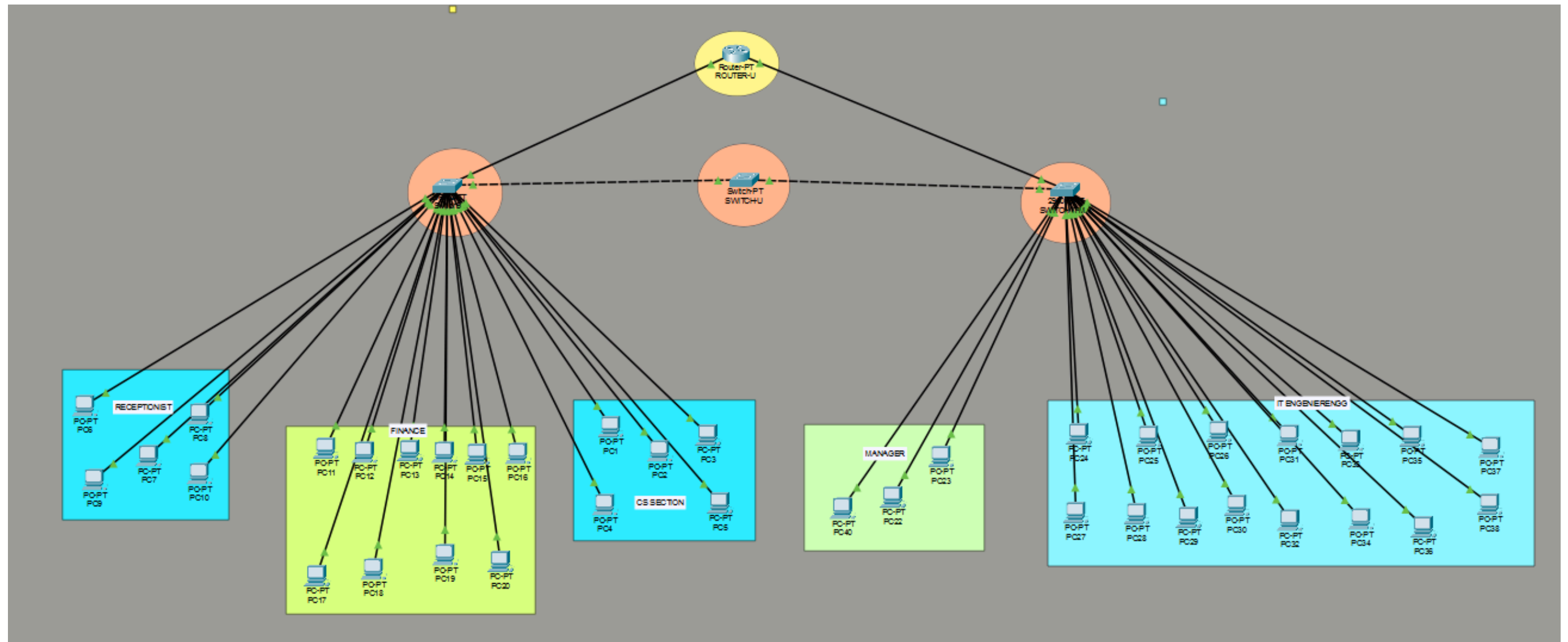
5. ENTERPRISE NETWORK DASHBOARD: VLAN & COST ANALYSIS

VLAN ID	JUMLAH PC
	5
	5
	10
	3
	15

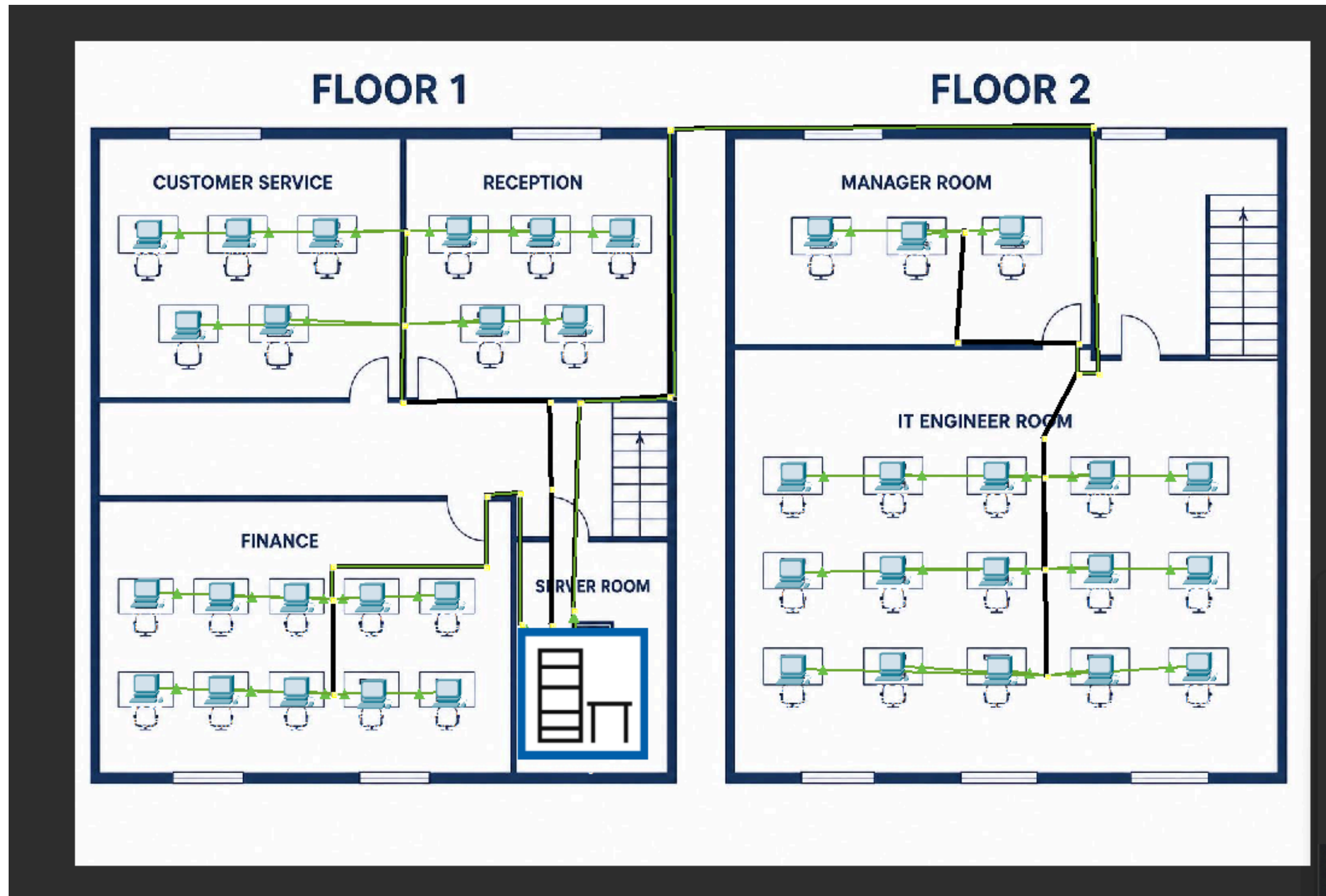
	Quantity Needed	Unit Price	Total price
PL SWITCH	38	Rp 1.500.000	Rp 43.700.000
	2		
	1	R 1.100.00	R 1.100.00
ROUTER	1	p 0	0
		R 3.940.00	R 3.940.00
		p 0	0



LOGICAL VIEW BY *CISCO PACKET TRACER*



PHYSICAL VIEW BY *CISCO PACKET TRACER*



CLIENT BRIEF (FIKTIF)



sarah lockhart <sarahlockhartparker@gmail.com>

kepada saya ▾

13.53 (22 menit yang lalu)



Halo Keysa,

Menindaklanjuti rencana pembukaan kantor cabang baru, kami membutuhkan bantuanmu untuk merancang infrastruktur jaringan lokal (LAN) yang mencakup 2 lantai dengan total sekitar 35+ endpoint.

Berikut adalah beberapa spesifikasi wajib yang harus dipenuhi:

1. Segmentasi Jaringan: Tolong pisahkan jaringan menjadi beberapa divisi (VLAN 10-50). Keamanan data divisi FINANCE (VLAN 30) harus diperketat maksimal karena mengelola data aset keuangan penting perusahaan.
2. Otomasi Port Security: Aktifkan pengamanan port pada switch Finance agar otomatis mati jika ada perangkat asing yang mencoba terhubung tanpa izin.
3. Otomasi IP Address: Semua komputer karyawan harus mendapatkan IP otomatis via DHCP agar tim operasional tidak perlu setting manual satu-per-satu.
4. Perangkat Hardening: Semua jalur akses masuk ke Switch/Router (Console, Enable, Remote VTY) wajib dikunci dengan password dan di-enkripsi menyeluruh agar tidak bisa di-intip dalam mode plain-text. Pasang juga Banner MOTD sebagai peringatan hukum akses ilegal.

Kami juga meminta rincian Bill of Materials (BoQ) dalam bentuk file Excel interaktif yang menampilkan diagram alokasi budget (CAPEX) agar manajemen bisa melihat total pengeluaran secara visual.

Kirimkan dokumentasi teknis beserta file simulasi Cisco Packet Tracer jika proyek ini sudah selesai di-audit.

Terima kasih atas kerja samanya.

Salam,
Sarah Parker,
Lead IT Infrastructure & Network Lecturer